

EE4520

Analog CMOS Design I

Week Q2.1
November 14th, 2024

The Team

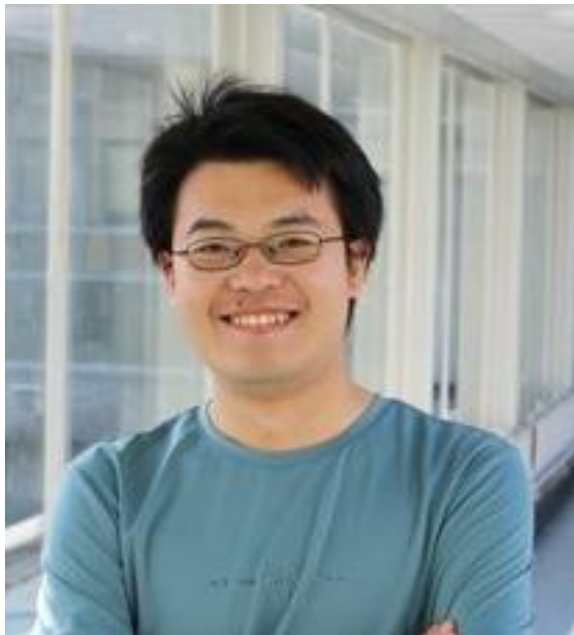


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Analog CMOS Design I

Analog

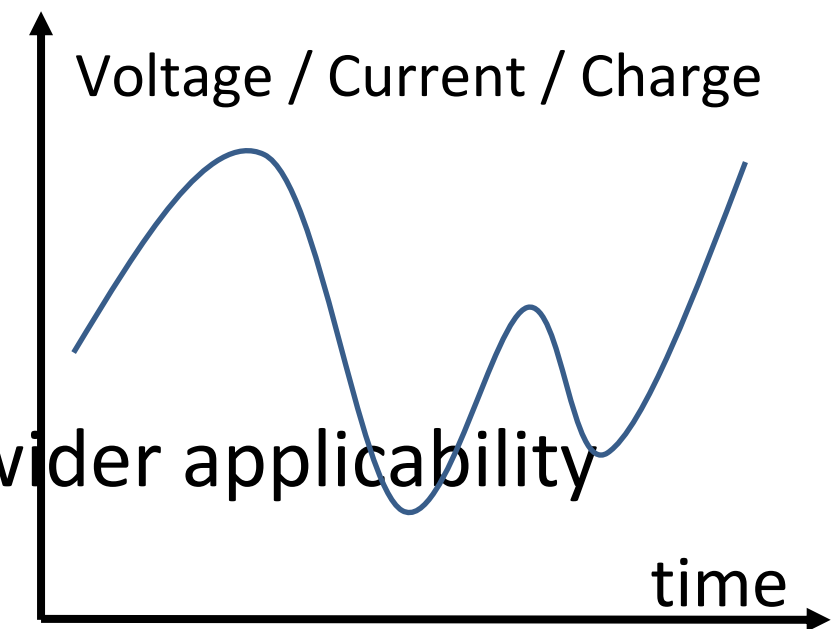
- Signals of interest (currents, voltages, charges) are continuous in time **and** amplitude

CMOS

- Focus on most used technology
- Although techniques and approach have wider applicability

Design

- Focus on circuit design!
- We assume you know the basics



I

- We are going to also have a more advanced part II

Prerequisites

- Basic device physics
- Circuit theory and circuit analysis
- **Basic circuits:** single-transistor amplifiers, differential pairs, current mirrors, biasing
- **Circuit imperfections:** frequency response, noise
- **Design/analysis technique**
- **Simulations of circuits:** transients, DC, AC analysis,...
- In short, all the content of EE4C10 (Analog Circuit Design Fundamentals)

What will you learn?

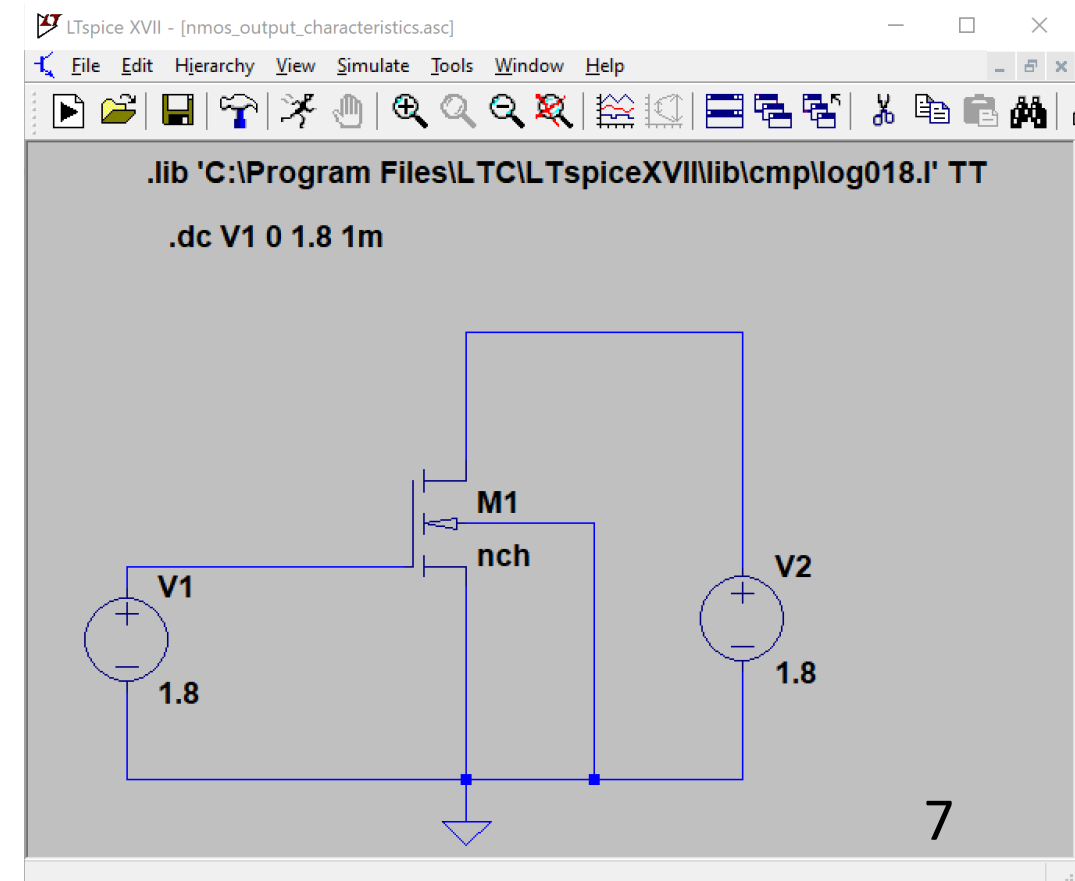
- Understand the role of MOS transistors in circuits, including second-order effects.
- Derive the performance of simple CMOS circuits (such as single-stage amplifiers)
- **Design and optimize simple CMOS circuits** (single-stage amplifiers, cased amplifiers).
- Derive performance of CMOS circuits by means of a Spice-like circuit simulator

Assessment

- HW1 – Deadline: December 20th, 2025 – 15%
- HW2 – Deadline: February 1st, 2026 – 85%
 - Grade determined by *quantitative* performance (65% of final grade)...
 - ...and by the design choices, to be described in a 1-page report (20% of final grade)
- **All assignments must be executed individually**
 - But you are allowed to discuss
- **Make good use of office hours with TAs**
- **Homework content:** design and optimize a folded-cascode amplifier

The simulator

- **LTspice XVII** from Analog Devices (previously Linear Technologies)
 - SPICE based simulator
 - Main features of commercial simulators
 - **Can run on your laptop**
 - **Freeware**
- Used for home assignment and in Thursday's class
- Download and install from [here](#)
- Detailed instruction on Brightspace
- A model file for 0.18- μm CMOS technology available in Brightspace



Course material

- On Brightspace
 - Slides pdf's
 - Software for simulation, device models, examples
 - Additional material when needed (papers, manuals,...)
- Reference material
 - Klaas Bult, *“Basic Analog CMOS Design, An Intuitive Approach”*
 - To be downloaded on Brightspace